

Week 1 Weekend Assignment

"Network Detective Challenge"

TheBigPicture

Over the past 4 days you learned:

- Day 1: What cybersecurity is — CIA Triad, AAA, threats vs vulnerabilities vs risks
- Day 2: Networking basics — OSI model, TCP/IP, IP addressing, ports, protocols
- Day 3: Common protocols — HTTP/HTTPS, FTP/SFTP, SSH, SMTP, DNS, ICMP + subnetting
- Day 4: Linux & Windows security — permissions, users, hardening, event logs

Now it is time to combine all of that into one real-world mini-project.

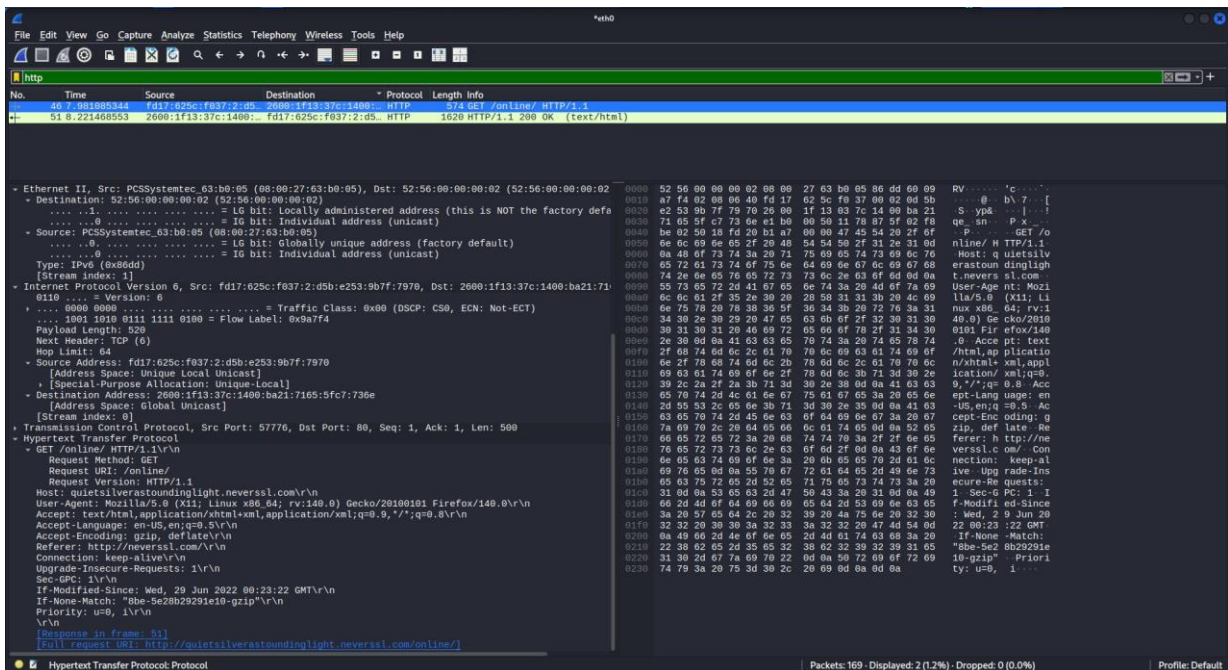
This assignment has 3 parts. Do all 3. They build on each other.

Part1—Capture&AnalyzeHTTPTraffic(Day2+Day3skills)

What to do:

1. Install Wireshark (if not already): <https://www.wireshark.org/download.html>
2. Open Wireshark and start capturing on your active network interface (Wi-Fi or Ethernet).
3. While capturing, open a browser and visit 3 different websites:
 - One that uses HTTP (try <http://neverssl.com> — it deliberately uses HTTP)
 - One that uses HTTPS (try <https://www.google.com>)
 - One of your choice (any site you normally use)
4. Stop the capture after about 60 seconds of browsing.
5. Apply the display filter http in Wireshark — this shows only HTTP traffic.
6. Also try the filter dns to see DNS queries.
7. Also try the filter tcp.port == 443 to see HTTPS traffic.

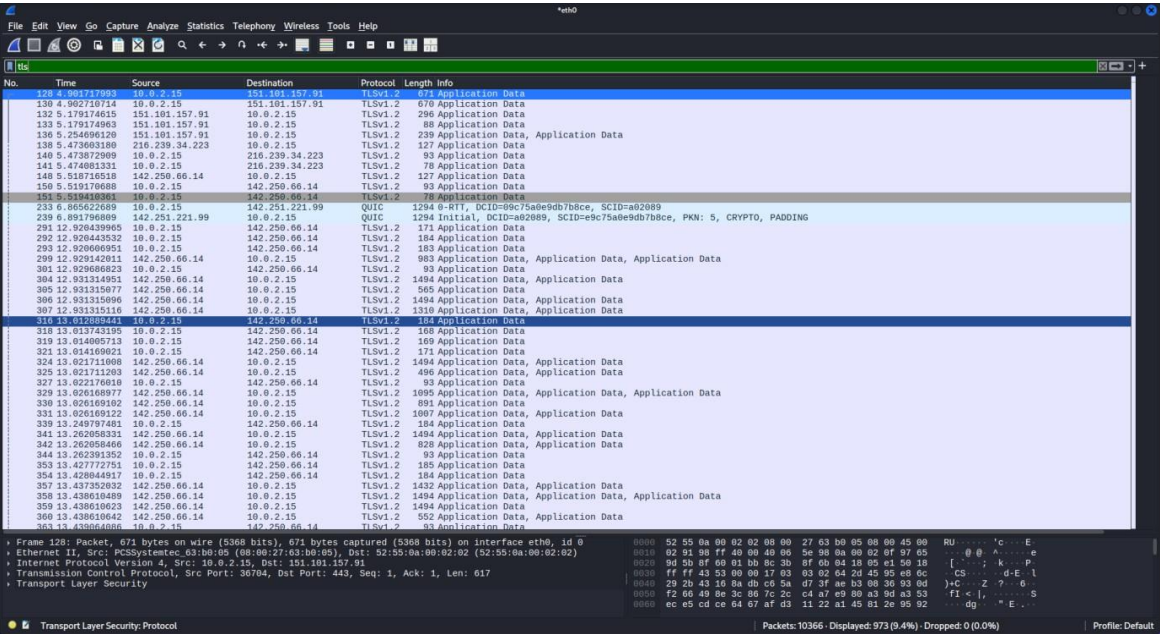
<http://neverssl.com>



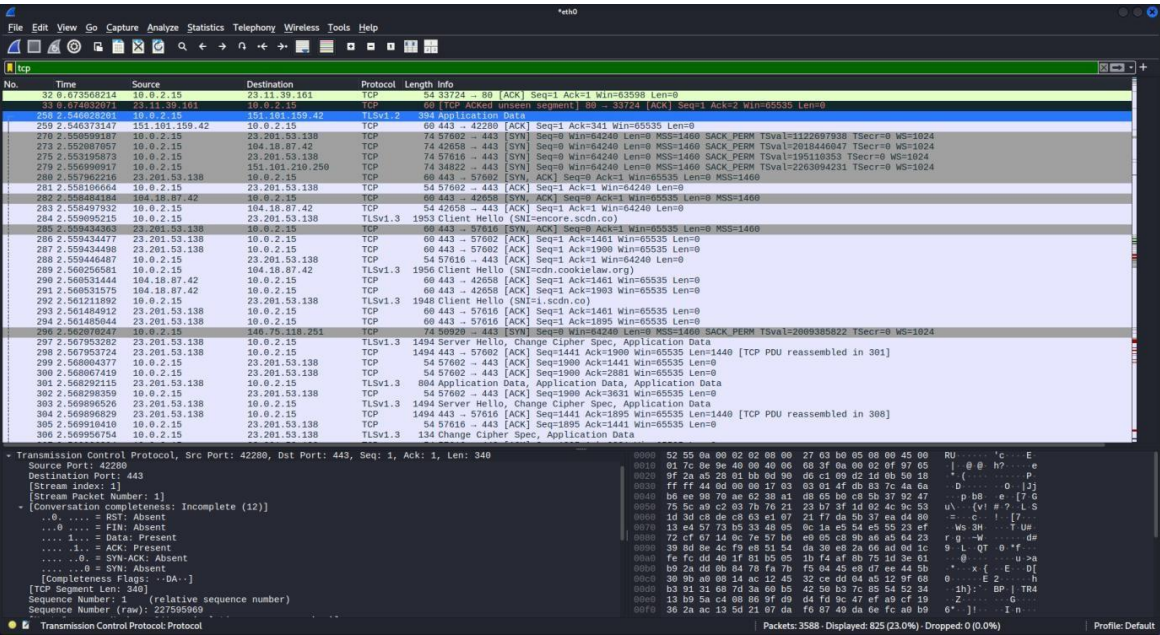
https://google.com

HTTPS encrypts all application-layer data. Wireshark still captures the raw network packets, but it labels them as TLS or TCP instead of HTTP

Protocol Labeling: Wireshark classifies packets based on what it can parse. Because the HTTP payload is entirely encrypted inside a TLS wrapper, Wireshark only identifies the underlying transport and security layers (TLSv1.2, TLSv1.3, or TCP)



https://open.spotify.com/



dns

No.	Time	Source	Destination	Protocol	Length	Info
244	2.539176117	10.0.2.15	192.168.1.1	DNS	79	Standard query 0xc7b8 HTTPS open.spotifycdn.com
245	2.539413189	10.0.2.15	192.168.1.1	DNS	74	Standard query 0x9056 HTTPS encore.scdn.co
246	2.539415348	10.0.2.15	192.168.1.1	DNS	78	Standard query 0x0296 A open.spotifycdn.com
247	2.539459663	10.0.2.15	192.168.1.1	DNS	77	Standard query 0xb1bc HTTPS cdn.cookiecaw.org
248	2.539460111	10.0.2.15	192.168.1.1	DNS	74	Standard query 0x6644 A encore.scdn.co
249	2.539460437	10.0.2.15	192.168.1.1	DNS	74	Standard query 0x5047 AAAA encore.scdn.co
250	2.539517189	10.0.2.15	192.168.1.1	DNS	69	Standard query 0xc4ab HTTPS i.scdn.co
251	2.539518011	10.0.2.15	192.168.1.1	DNS	77	Standard query 0xb698 A cdn.cookiecaw.org
252	2.539518138	10.0.2.15	192.168.1.1	DNS	77	Standard query 0xf0f9 AAAA cdn.cookiecaw.org
253	2.539518669	10.0.2.15	192.168.1.1	DNS	69	Standard query 0x4b62 A i.scdn.co
254	2.539518888	10.0.2.15	192.168.1.1	DNS	69	Standard query 0xd6c2 AAAA i.scdn.co
255	2.539588076	10.0.2.15	192.168.1.1	DNS	83	Standard query 0xc266 HTTPS pickasso.spotifycdn.com
256	2.540116154	10.0.2.15	192.168.1.1	DNS	83	Standard query 0xc35d A pickasso.spotifycdn.com
257	2.540373492	10.0.2.15	192.168.1.1	DNS	18	Standard query 0x10c7 HTTPS global.tls13.sp
258	2.540373492	192.168.1.1	10.0.2.15	DNS	186	Standard query response 0xc7b8 HTTPS open.spotifycdn.com CNAME tls13brtt.spotifycdn.map.fastly.net SOA ns1.fastly.net
262	2.548188932	192.168.1.1	10.0.2.15	DNS	389	Standard query response 0x9056 HTTPS encore.scdn.co CNAME encore.spotifycdn.com CNAME default.cdn-gslb.spotify.com CNAME common
263	2.548181136	192.168.1.1	10.0.2.15	DNS	355	Standard query response 0xc4ab HTTPS i.scdn.co CNAME image-scdn.cdn-gslb.spotify.com CNAME common-elip-ak.spotifycdn.com edgesuite
264	2.548181157	192.168.1.1	10.0.2.15	DNS	368	Standard query response 0x6644 A encore.scdn.co CNAME encore.spotifycdn.com CNAME default.cdn-gslb.spotify.com CNAME common-elip
265	2.548181183	192.168.1.1	10.0.2.15	DNS	194	Standard query response 0xb1bc HTTPS cdn.cookiecaw.org HTTPS
266	2.548181203	192.168.1.1	10.0.2.15	DNS	216	Standard query response 0x296 A open.spotifycdn.com CNAME tls13brtt.spotifycdn.map.fastly.net A 146.75.118.251 NS ns1.fastly.net
267	2.548387239	192.168.1.1	10.0.2.15	DNS	78	Standard query 0xc357 AAAA open.spotifycdn.com
268	2.549176111	192.168.1.1	10.0.2.15	DNS	384	Standard query response 0x5047 AAAA encore.scdn.co CNAME encore.spotifycdn.com CNAME default.cdn-gslb.spotify.com CNAME common
269	2.550261978	192.168.1.1	10.0.2.15	DNS	195	Standard query response 0xf0f9 AAAA cdn.cookiecaw.org AAAA 2686:4706:83b3:3663:e7fd:797:6812:562a
271	2.551295169	192.168.1.1	10.0.2.15	DNS	189	Standard query response 0xb698 A cdn.cookiecaw.org A 184.18.87.42 A 184.18.86.42
272	2.551295985	192.168.1.1	10.0.2.15	DNS	326	Standard query response 0x4b62 A i.scdn.co CNAME image-scdn.cdn-gslb.spotify.com CNAME common-elip-ak.spotifycdn.com edgesuite
274	2.552252502	192.168.1.1	10.0.2.15	DNS	350	Standard query response 0xd6c2 AAAA i.scdn.co CNAME image-scdn.cdn-gslb.spotify.com CNAME common-elip-ak.spotifycdn.com edgesuite
276	2.555290278	192.168.1.1	10.0.2.15	DNS	246	Standard query response 0xc266 HTTPS pickasso.spotifycdn.com CNAME region-eng-spotifycdn.cdn-gslb.spotify.com CNAME global.tls13.sp
277	2.556511225	192.168.1.1	10.0.2.15	DNS	276	Standard query response 0xc35d A pickasso.spotifycdn.com CNAME region-eng-spotifycdn.cdn-gslb.spotify.com CNAME global.tls13.sp
278	2.556511583	192.168.1.1	10.0.2.15	DNS	216	Standard query response 0xf0f9 AAAA pickasso.spotifycdn.com CNAME region-eng-spotifycdn.cdn-gslb.spotify.com CNAME global.tls13.sp
295	2.561090933	192.168.1.1	10.0.2.15	DNS	228	Standard query response 0xc357 AAAA open.spotifycdn.com CNAME tls13brtt.spotifycdn.map.fastly.net AAAA 2686:4706:83b3:3663:e7fd:797:6812:562a
383	2.796093633	10.0.2.15	192.168.1.1	DNS	78	Standard query 0x270e A pl.scdn.co
384	2.796093633	10.0.2.15	192.168.1.1	DNS	78	Standard query 0x10d0 AAAA pl.scdn.co
385	2.796093645	10.0.2.15	192.168.1.1	DNS	163	Standard query response 0x270e No such name AAAA pl.scdn.co SOA ns-cloud-d1.googledomains.com
386	2.796460834	192.168.1.1	10.0.2.15	DNS	163	Standard query response 0x10d0 No such name AAAA pl.scdn.co SOA ns-cloud-d1.googledomains.com

Frame 257: Packet, 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface eth0, id 0

Ethernet II, Src: PCSysmetec, 63:b0:05 (08:00:27:63:b0:05), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 192.168.1.1

User Datagram Protocol, Src Port: 48013, Dst Port: 53

Source Port: 48013

Destination Port: 53

Length: 49

Checksum: 0xcdfa [unverified]

Checksum Status: Unverified

Stream Index: 11

Stream Packet Number: 2

Timestamps

Time since first frame: 149.952 microseconds

Time since previous frame: 149.952 microseconds

UDP payload (41 bytes)

User Datagram Protocol (query)

Domain Name System (query)

Domain Name System: Protocol

Packets: 3588 - Displayed: 44 (1.2%) - Dropped: 0 (0.0%) Profile: Default

DNS (Domain Name System) is the phonebook of the Internet, responsible for translating human-readable domain names into machine-readable numeric IP addresses.

No.	Time	Source	Destination	Protocol	Length	Info
258	2.546028201	10.0.2.15	151.101.159.42	TLSv1.2	394	Application Data
259	2.546033147	151.101.159.42	10.0.2.15	TCP	60	443 -> 42280 [ACK] Seq=1 Ack=341 Win=65535 Len=0
278	2.550878857	10.0.2.15	23.201.53.138	TCP	74	57602 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1122697938 TSecr=0 WS=1024
279	2.550878857	10.0.2.15	23.201.53.138	TCP	74	42058 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2018446047 TSecr=0 WS=1024
275	2.553195873	10.0.2.15	23.201.53.138	TCP	74	57616 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=195118353 TSecr=0 WS=1024
276	2.550999917	10.0.2.15	151.101.210.250	TCP	74	34822 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2263094231 TSecr=0 WS=1024
280	2.551962215	10.0.2.15	23.201.53.138	TCP	60	443 -> 57602 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
281	2.558106664	10.0.2.15	23.201.53.138	TCP	54	57602 -> 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
282	2.55848184	10.0.2.15	10.0.2.15	TCP	60	443 -> 42658 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
283	2.558497932	10.0.2.15	10.0.2.15	TCP	54	42658 -> 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
284	2.558995215	10.0.2.15	23.201.53.138	TLSv1.3	1953	Client Hello (SNI=encore.scdn.co)
285	2.559434363	23.201.53.138	10.0.2.15	TCP	60	443 -> 57616 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
286	2.559434477	23.201.53.138	10.0.2.15	TCP	60	443 -> 57602 [ACK] Seq=1 Ack=1461 Win=65535 Len=0
287	2.559434498	23.201.53.138	10.0.2.15	TCP	60	443 -> 57602 [ACK] Seq=1 Ack=1908 Win=65535 Len=0
288	2.559446487	10.0.2.15	23.201.53.138	TCP	54	57616 -> 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
289	2.560256581	10.0.2.15	10.0.2.15	TLSv1.3	1956	Client Hello (SNI=cdn.cookiecaw.org)
290	2.560314444	10.0.2.15	10.0.2.15	TCP	60	443 -> 42658 [ACK] Seq=1 Ack=1461 Win=65535 Len=0
291	2.560315175	10.0.2.15	10.0.2.15	TCP	60	443 -> 42658 [ACK] Seq=1 Ack=1903 Win=65535 Len=0
292	2.561211092	10.0.2.15	23.201.53.138	TLSv1.3	1948	Client Hello (SNI=i.scdn.co)
293	2.561484912	23.201.53.138	10.0.2.15	TCP	60	443 -> 57616 [ACK] Seq=1 Ack=1461 Win=65535 Len=0
294	2.561485044	23.201.53.138	10.0.2.15	TCP	60	443 -> 57616 [ACK] Seq=1 Ack=1895 Win=65535 Len=0
296	2.568078247	10.0.2.15	146.75.118.251	TCP	74	59928 -> 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2009385022 TSecr=0 WS=1024
297	2.567953282	23.201.53.138	10.0.2.15	TLSv1.3	1494	Server Hello, Change Cipher Spec, Application Data
298	2.567953724	23.201.53.138	10.0.2.15	TCP	1494	443 -> 57602 [ACK] Seq=1441 Ack=1900 Win=65535 Len=1440 [TCP PDU reassembled in 301]
299	2.568084377	10.0.2.15	23.201.53.138	TCP	54	57602 -> 443 [ACK] Seq=1900 Ack=1441 Win=65535 Len=0
300	2.568067419	10.0.2.15	23.201.53.138	TCP	54	57602 -> 443 [ACK] Seq=1900 Ack=2881 Win=65535 Len=0
301	2.568292115	23.201.53.138	10.0.2.15	TLSv1.3	884	Application Data, Application Data, Application Data
302	2.568298359	10.0.2.15	23.201.53.138	TCP	54	57602 -> 443 [ACK] Seq=1900 Ack=3631 Win=65535 Len=0
303	2.568996526	23.201.53.138	10.0.2.15	TLSv1.3	1494	Server Hello, Change Cipher Spec, Application Data
304	2.568996829	23.201.53.138	10.0.2.15	TCP	1494	443 -> 57616 [ACK] Seq=1441 Ack=1895 Win=65535 Len=1440 [TCP PDU reassembled in 308]
305	2.569918410	10.0.2.15	23.201.53.138	TCP	54	57616 -> 443 [ACK] Seq=1895 Ack=1441 Win=65535 Len=0
306	2.569956754	10.0.2.15	23.201.53.138	TLSv1.3	134	Change Cipher Spec, Application Data
307	2.569989389	10.0.2.15	23.201.53.138	TCP	54	57616 -> 443 [ACK] Seq=1895 Ack=2383 Win=65535 Len=0
308	2.570187486	23.201.53.138	10.0.2.15	TLSv1.3	885	Application Data, Application Data, Application Data

Protocols in frame: eth:ethertype:ip:tcp:tls

Character encoding: ASCII (0)

Coloring Rule Name: TCP

Coloring Rule String: tcp

Ethernet II, Src: PCSysmetec, 63:b0:05 (08:00:27:63:b0:05), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)

Destination: 52:55:0a:00:02:02 (52:55:0a:00:02:02)

Source: PCSysmetec, 63:b0:05 (08:00:27:63:b0:05)

Type: IPv4 (0x8006)

Type: IPv4 (0x8006)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 151.101.159.42

0100 ... = Version: 4

0101 ... = Header Length: 20 bytes (5)

wireshark_eth0AEROQ3.pcapng

Packets: 3588 - Displayed: 767 (21.4%) - Dropped: 0 (0.0%) Profile: Default

tcp.port == 443

TCP port 443 is the standard, default port for **HTTPS** (Hypertext Transfer Protocol Secure) traffic. It enables secure, encrypted communication between a client (like your web browser) and a web server, protecting sensitive data during

activities like online banking, e-commerce, and login authentications.

What to write about (answer in your own words):

For each of the 3 websites, answer:

Question	Your answer
What website did you visit?	open.spotify.com
HTTP or HTTPS? How can you tell from the capture?	HTTPS. Evidence in the capture: The Protocol column shows TLSv1.2 / TLSv1.3, not HTTP
What IP address did your computer connect to?	10.0.2.15
What was the destination port? (80 or 443?)	443
For HTTP sites: what information is visible in the packet?	URLs, query parameters, headers, cookies, and page content.
For HTTPS sites: what information is visible vs hidden?	Visible: The website's domain name (like encore.scdn.co found in the Client Hello SNI field) and the network routing data. Hidden: The specific pages visited, form inputs, passwords, and the actual content (hidden inside the encrypted Application Data blocks).
What DNS queries did you see before the connection?	A/AAAA for open.spotifycdn.com → CNAME tls130rtt.spotifycdn.map.fastly.net → A 146.75.118.251

Key concepts to connect:

- CIA Triad: When you visited the HTTP site, which part of CIA was violated? (Hint: no encryption = no Confidentiality)

The Confidentiality part of the CIA triad was violated.

- AAA: If the HTTP site had a login page, which AAA function would be exposed? (Hint: Authentication credentials in clear text)

The Authentication function would be exposed.

- Subnetting: What is the IP range of your home network? (Check with ipconfig or ifconfig) What subnet mask are you using?

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ifconfig  
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255  
    ether 62:ab:19:a3:c6:51 txqueuelen 0 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 6 overruns 0 carrier 0 collisions 0  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255  
    inet6 fd17:625c:f037:2:d5b:e253:9b7f:7970 prefixlen 64 scopeid 0x0<global>  
    inet6 fe80::512:59a3:b52e:3606 prefixlen 64 scopeid 0x20<link>  
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)  
    RX packets 79402 bytes 92149847 (87.8 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 28950 bytes 4461192 (4.2 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 10 bytes 580 (580.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 10 bytes 580 (580.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$
```

Subnet Mask : 255.255.255.0

IP Address: 10.0.2.15

IP Range: 10.0.2.0 to 10.0.2.255

Where to get help / sources:

- Wireshark official docs: <https://www.wireshark.org/docs/>
- Wireshark 101 tutorial: <https://www.youtube.com/watch?v=TkCSr30Uo>

How to filter HTTP traffic: In Wireshark, type http in the display filter bar

Part2—SubnettingPractice(Day3skills)

Using what you learned about subnetting in Day 3, solve these problems. Show your working.

Problem 1

Your home router gives you the network 192.168.1.0/24.

- How many total host addresses are available in this network?

$2^8 = 256$ total IP addresses. subtracting 2 addresses for the network ID and the broadcast address ($256 - 2$), there are 254 usable host addresses.

- If you split it into 4 equal subnets, what is the new subnet mask in CIDR notation?

To split a network into 4 equal subnets, you need to borrow additional bits from the host portion ($2^2 = 4$) Adding 2 bits to the original mask gives the new CIDR notation: **$/24 + 2 = /26$**

- What are the network addresses of all 4 subnets?

Each **$/26$** subnet increments by **$2^{32-26} = 2^6 = 64$** addresses:

Subnet 1: 192.168.1.0

Subnet 2: 192.168.1.64

Subnet 3: 192.168.1.128

Subnet 4: 192.168.1.192

- What is the broadcast address of the third subnet?

192.168.1.191

- How many usable hosts per subnet after splitting?

Each **$/26$** subnet contains a total of **$2^6 = 64$** , Subtracting the network address & broadcast address (**$64 - 2$**) **62 usable hosts** per subnet.

Problem 2

You are a network admin for a small office. You have these departments:

- Sales: needs 30 hosts
- Engineering: needs 50 hosts
- HR: needs 10 hosts
- Servers: needs 10 hosts

You are given the network 10.0.0.0/24.

- Design a subnetting scheme that gives each department enough addresses.

1. Engineering (50 hosts):

- Network: 10.0.0.0/26
- Usable IPs: 10.0.0.1 to 10.0.0.62
- Broadcast: 10.0.0.63

2. Sales (30 hosts):

- Network: 10.0.0.64/27
- Usable IPs: 10.0.0.65 to 10.0.0.94
- Broadcast: 10.0.0.95

3. HR (10 hosts):

- Network: 10.0.0.96/28
- Usable IPs: 10.0.0.97 to 10.0.0.110
- Broadcast: 10.0.0.111

4. Servers (10 hosts):

- Network: 10.0.0.112/28
- Usable IPs: 10.0.0.113 to 10.0.0.126
- Broadcast: 10.0.0.127

(Addresses from 10.0.0.128 to 10.0.0.255 remain completely free for future growth).

- For each department, write: network address, subnet mask (CIDR), usable host range, broadcast address.

1. Engineering

- **Network Address:** 10.0.0.0
- **Subnet Mask (CIDR):** /26
- **Usable Host Range:** 10.0.0.1 – 10.0.0.62
- **Broadcast Address:** 10.0.0.63

2. Sales

- **Network Address:** 10.0.0.64
- **Subnet Mask (CIDR):** /27
- **Usable Host Range:** 10.0.0.65 – 10.0.0.94
- **Broadcast Address:** 10.0.0.95

3. HR

- **Network Address:** 10.0.0.96
- **Subnet Mask (CIDR):** /28
- **Usable Host Range:** 10.0.0.97 – 10.0.0.110
- **Broadcast Address:** 10.0.0.111

4. Servers

- **Network Address:** 10.0.0.112
- **Subnet Mask (CIDR):** /28
- **Usable Host Range:** 10.0.0.113 – 10.0.0.126
- **Broadcast Address:** 10.0.0.127

- Which department gets the largest subnet? Why?

Engineering gets the largest subnet.

Why: It has the highest host requirement (50 hosts). In networking, block sizes must scale up in powers of 2. The closest power of 2 that safely covers 50 hosts (plus the 2 mandatory routing addresses for the network ID and broadcast) is $2^6 = 64$. This requires a /26 mask, which consumes a larger share of the IP pool than any other department.

Problem 3 — Connect to Security

- Why would a security team want to put the Servers department on a different subnet from Sales? (Answer using CIA + AAA concepts)

Confidentiality & Integrity (CIA): If a sales computer gets infected with malware or hacked, it can easily infect everything on its own subnet. Separating the Servers puts them behind a virtual wall, protecting company data from being stolen (**Confidentiality**) or deleted (**Integrity**).

Access Control (AAA): It lets you place a firewall between the two departments. This means you can track exactly who is trying to connect (**Authentication**), block Sales from accessing parts of the servers they don't need (**Authorization**), and log all traffic to spot suspicious behaviour (**Accounting**).

- If an attacker compromises a Sales PC, can they directly access a Server on a different subnet? Explain why or why not.

No, they cannot directly access it. Because they are on different subnets, traffic cannot just flow freely between them. Here is exactly why:

They have to go through a Router: Devices on the same subnet can talk directly to each other. But to talk to a different subnet, the attacker's traffic *must* be sent to the local router to find a path to the Server network.

The Router/Firewall acts as a Guard: Because the traffic has to pass through the router, the security

team can place a firewall right at that boundary. This firewall blocks unauthorized traffic by default.

Unless the security team explicitly created a rule allowing that specific Sales PC to talk to that specific Server port, the firewall will drop the attacker's connection attempt immediately.

- What device controls traffic between subnets? (Hint: think about Day 2 — routers/firewalls)
The Router (The Bridge): Devices on different subnets cannot talk to each other directly. The router acts as the "default gateway." It reads the destination IP address and handles the actual steering of data from one subnet to another.
The Firewall (The Security Guard): While the router connects the subnets, the firewall decides *if* the traffic is allowed to pass. It sits on the boundary and checks every packet against security rules to block unauthorized access (like stopping the Sales subnet from reaching the Server subnet).

Part3—ConnectEverything:CIA+AAAAcrossAll4Days

This is the thinking part. Write short paragraphs (3-5 sentences each). No copy-paste — use your own words.

Question 1 — CIA Triad in Action

Pick one of the protocols you studied (HTTP, FTP, SMTP, DNS, SSH, or ICMP). For that protocol:

- How does it support one part of the CIA Triad? HTTP supports **Availability**
- How does it fail to protect one part of the CIA Triad? HTTP completely fails to protect **Confidentiality**.
- What would a security professional do to fix that weakness? transition the site to **HTTPS** by installing an SSL/TLS certificate on the web server.

Question 2 — AAA Framework in Action

Think about logging into a website (like Gmail or any site you use).

- Authentication: How does the site verify you are who you say you are? **Login id and passwords**
- Authorization: After you log in, what are you allowed to do? What are you NOT allowed to do? Check **access control list** and allow when access is granted and denied no access is granted.
- Accounting: What records might the site keep about your login? Why?

The exact date and timestamp, IP address, geographic location, and the specific device or browser you are using, instantly flag unusual behavior—like a sudden login from another country—and provide an audit trail to figure out exactly what went wrong if your account ever gets compromised.

Now answer: What happens if an attacker captures your HTTP traffic from Part 1? Which part of AAA breaks first? Why?

Question 3 — The Full Attack Story

Imagine this scenario:

An employee at a company uses an unencrypted HTTP connection to log into the internal HR portal from a coffee shop. The company's network is poorly segmented — all departments are on the same subnet 192.168.1.0/24.

Using everything you learned this week, write a short attack story:

1. What can the attacker on the coffee shop Wi-Fi see? (Day 3 — HTTP, packet capture)
The attacker sees the employee's raw, unencrypted username and password in plain text.
2. What can the attacker do with those credentials? (Day 1 — Authentication broken, CIA violated)
The attacker bypasses authentication to steal private HR data and alter payroll records.
3. Once inside the company network, what can the attacker reach? (Day 3 — subnetting, no segmentation)
The attacker can reach every server and computer in the company due to the lack of internal walls.
4. How could the company have prevented this? (Day 4 — hardening, Day 3 — segmentation, Day 1 — controls)
The company should have enforced HTTPS encryption, multi-factor authentication, and firewall segmentation.

Question 4 — Linux/Windows Connection

- On Day 4 you learned about file permissions (Linux) and Event Viewer (Windows).

File permission using chmod and rwx,421 combination, Event Viewer for Windows Logs to investigate(soc).

- How does the principle of least privilege (Day 4) connect to the AAA Authorization function (Day 1)?

Authorization is the tool used to enforce Least Privilege.

- If an attacker gets a shell on a Linux machine, what permission-related files would they look for?

/etc/shadow and sudo privilege

- On Windows, where would you look in Event Viewer to spot a brute-force login attempt?

Event ID 4625 logs

HowtoSubmit

Create a single document (text file, markdown, or PDF) with:

1. Part 1: Your traffic analysis table + answers + at least 2 screenshots from Wireshark
2. Part 2: All subnetting answers with working shown
3. Part 3: All 4 written answers in your own words

Name it: Week1_Weekend_Assignment_YourName.md (or .txt or .pdf)

Grading Rubric (self-grade before submitting)

Criteria	Points	Your score
Part 1: Traffic captured and analyzed correctly	/25	
Part 1: CIA + AAA connections made	/10	
Part 2: Subnetting calculations correct	/25	
Part 2: Security reasoning about subnets	/10	
Part 3: Own words, no copy-paste	/15	
Part 3: Connections across all 4 days are accurate	/15	
Total	/100	

TipsforDoingThisWell

- Do Part 1 first. It is the most hands-on and will make the theory in Parts 2 and 3 feel real.
- Use your notes from all 4 days. This assignment is designed to make you go back and review everything.
- If you get stuck on subnetting, re-read the Day 3 notes. The 192.168.1.0/24 example is your starting point.
- Screenshots matter. In cybersecurity, evidence matters. Show what you captured.
- Write like you are explaining to a friend. If you cannot explain it simply, you do not understand it yet.